worsening of vision subjectively. There were an additional 8 patients (7.8%) with slight worsening of visual acuity, 2 patients (1.9%) with minor loss of visual field, and 2 patients (1.9%) with subtle losses of both visual acuity and visual field. Among patients with either subjectively or objectively worse vision, 12 of 39 patients (30.8%) had known tumor recurrence, 7 patients (17.9%) had exposure keratopathy, and 8 patients (20.5%) had other known cause of decreased vision (2 cataracts, 1 diabetic retinopathy, 1 glaucoma, 1 posterior vitreous detachment, 1 epidermoid cyst of the eye, 1 keratoconus, 1 papilledema). Four of the 103 patients (3.9%) developed RION and all had a baseline visual acuity of 16/20 or better prior to irradiation. Median time to develop RION was 27.5 months (range: 7-110 months). Maximum radiation dose to the chiasm or optic nerve affected was greater than 60 Gy (RBE) in all cases. However, the incidence of RION was only 4.8% among 83 patients receiving $\geq$ 60 Gy (RBE) to the optic chiasm or either optic nerve.

Conclusions: After proton radiation, a significant number of skull base chordoma and chondrosarcoma patients may experience worsening vision attributable to disease recurrence and other unrelated conditions. RION remains a low but real risk among patients receiving $\geq$ 60 Gy (RBE) to the optic pathway. Some patients can experience vision improvement following radiation therapy.

Author Disclosure: **P.C. Li:** None. **N.J. Liebsch:** None. **A. Niemierko:** None. **H.A. Shih:** A. Employee; Dartmouth-Hitchcock Medical Center. F. Honoraria; IJROBP. H. Speakers Bureau; Merck. K. Advisory Board; Novartis. P. Royalty; UpToDate.

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Ruthenium-106 Plaque Brachytherapy for Uveal Melanoma: Factors Associated With Local Tumor Recurrence

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Purpose/Objective(s): Plaque brachytherapy is a common form of treatment for uveal melanoma, and the Collaborative Ocular Melanoma Study (COMS) employed I-125. Recently, Ru-106 has been reintroduced for plaque brachytherapy in the United States. We reviewed our experience treating uveal melanoma with Ru-106 plaque brachytherapy using COMS planning techniques, hypothesizing we would observe similar outcomes to those in the COMS.

Materials/Methods: Medical records of patients undergoing Ru-106 plaque brachytherapy were reviewed retrospectively. Patient, tumor, and treatment characteristics were recorded. Outcomes including visual acuity (VA), local tumor recurrence (LTR), salvage local treatment, metastasis and survival were recorded. Cox regression analyses were used to determine factors associated with local tumor recurrence and enucleation. Kaplan-Meier methods were used to estimate the frequency of events.

Results: Twenty-eight patients were studied. Median age was 60 and 50% were men. Median tumor base diameter and height were 9.4 and 2.6 mm. Median radiation dose delivered to tumor apex was 75.5 Gy. No patient developed neovascular glaucoma and VA $>$ 20/200 was preserved in 93.3% of patients not requiring salvage local therapy. With median follow-up of 71 months, LTR and enucleation occurred in 13 and 4 patients, for a 5-year estimated risk of LTR and enucleation of 41.5% (95% CI = 20.2-61.7%) and 16% (95% CI = 0.7-31.3%), respectively. All patients with LTR had salvage therapy. LTR was associated with low VA in the tumor-bearing eye (HR = 2.4, 95% CI = 1.7-3.1, $p = 0.02$), tumor proximity to optic disc (HR = 2.4, 95% CI = 1.5-2.4, $p = 0.005$), tumor proximity to foveal avascular zone (HR = 2.1, 95% CI = 1.7-2.5, $p = 0.0001$), posterior tumor border proximity to posterior pole (HR = 4.8, 95% CI = 3.6-6.0, $p = 0.009$), small plaque size (HR = 3.6, 95% CI = 2.4-4.8, $p = 0.04$) and difference in plaque and tumor diameter <6 mm (HR = 6.2, 95% CI = 4.7-7.8, $p = 0.02$). Enucleation was associated with low VA in tumor-bearing eye (HR = 4.1, 95% CI = 2.9-5.2, $p = 0.02$), low VA in fellow eye (HR = 11.2, 95% CI = 8.8-13.5, $p = 0.04$), and tumor proximity to foveal avascular zone (HR = 3.2, 95% CI = 2.2-4.2, $p = 0.02$). Gender, age, tumor apical height and basal diameter, retinal detachment over tumor, tumor shape, radiation dose at tumor apex or base, plaque serial number or adjuvant transpupillary thermotherapy were not associated with LTR or enucleation. Estimated 5-year rate of death and metastasis was 18.5% and 11.4%.

Conclusions: Among patients treated with Ru-106 plaque brachytherapy using COMS planning techniques, we found a greater than expected rate of LTR, but other outcomes similar to those observed in the COMS. These results suggest that Ru-106 plaque brachytherapy should be planned carefully at centers that have previously used COMS protocols and I-125.

Author Disclosure: **C.A. Barker:** A. Employee; Memorial Sloan-Kettering Cancer Center. E. Research Grant; ASTRO. G. Consultant; RP Pharmaceuticals, Nucletron. I. Travel Expenses; MASCC, Nucletron. S. Leadership; ASTRO. **J.H. Francis:** A. Employee; Memorial Sloan Kettering Cancer Center. **G.N. Cohen:** A. Employee; Memorial Sloan Kettering Cancer Center. G. Consultant; Nucletron. I. Travel Expenses; Nucletron. **B.P. Marr:** A. Employee; Memorial Sloan Kettering Cancer Center. **S.L. Wolden:** A. Employee; Memorial Sloan Kettering Cancer Center. **B. McCormick:** A. Employee; Memorial Sloan Kettering Cancer Center. **D.H. Abramson:** A. Employee; Memorial Sloan Kettering Cancer Center.

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Clinical Effect of UV Radiation on Ocular Melanoma: SEER Investigation

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Purpose/Objective(s): Ocular melanoma, with an incidence of 1 per 100,000 in western countries, is a rare but serious illness that leads to blindness and death. As the effects of ultraviolet (UV) radiation on cutaneous melanoma are further explored, the eye presents an opportunity to study both UV-exposed external & protected- internal primary locations in adjacent tissues. This study analyzes patients from the SEER database to determine how the UV environment affects stage and age at diagnosis, primary location, responsiveness to radiation and ultimately survival.

Materials/Methods: The SEER database provided 9077 ocular melanoma patients diagnosed from 1973 to 2010. The average annual UV-index for the state of birth was calculated, using data from the National Weather Service. Stage and age at diagnosis as well as primary location were trended with linear regression per increasing unit UV (PIUUV). Survival was analyzed with Kaplan-Meier, Log-Rank and Cox-Hazard Ratios (HR) for both overall survival and disease-specific survival.

Results: Conjunctival disease had an absolute 12% increase ($p < 0.001$) of local-stage disease and the more prevalent choroidal disease had a 1% decrease ($p = 0.04$) of local disease PIUUV. Age of diagnosis was 1 year younger PIUUV ($p < 0.001$) driven by younger diagnosis of local-disease patients ($p < 0.001$). Conjunctival disease was diagnosed 2.7 years ($p = 0.007$) younger and choroidal disease was diagnosed 0.8 years ($p = 0.002$) younger PIUUV. Changes in the UV-index did not cause any significant effect on the distribution of primary disease between sub-locations around the eye. Survival analysis found a correlation between greater UV-exposure and improved overall survival ($p < 0.001$). Cox HR analysis showed worse overall survival with advanced staging, location of retina, or overlapping with adjacent adnexa, bilateral disease, older or poorer patients and undergoing surgery and better outcomes with increased UV-exposure ($p = 0.038$). Disease specific HR analysis did not maintain this UV result. Increased UV-exposure correlated with improved outcomes for patients treated with radiation, both overall and disease-specific ($p = 0.01$ for both).

Conclusions: These results provide new and important data to better understand the pathogenesis of melanoma as a disease that evolves from both environmental and internal factors. The observed correlation between increased UV-exposure and better radiation treatment outcomes suggests a factor that prospective studies can further probe and improve future outcomes.

Author Disclosure: **S. Szeja:** None. **J.C. Jo:** None. **E. Xanthopoulos:** None. **K. Hager:** None. **J.K. March:** None. **S.V. Dandapani:** None.